GEMS ARTS & SCIENCE COLLEGE (AUTONOMOUS)

(Global Education And Management Studies)
Affiliated to University of Calicut

PROJECT REPORT

Entitled on

SMART EDUCONNECT

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2023-2026

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Valued on: .....

INTERNAL EXAMINER **EXTERNAL EXAMINER**

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Signature of the Guide

Head of the Department

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During this period they were found to be hardworking, punctual & efficient. We wish them success in their future endeavors.

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ACKNOWLEDGEMENT

I thank the almighty for giving me the courage and the perseverance in completing the project. This project itself is an acknowledgement for all those who have given as their heart- felt-co-operation in making it a grand success.

I am thankful to our principal, **Dr. Naveen Mohan** for providing the necessary infrastructure and labs. We are greatly indebted to Head of computer science , **Mr. Anoos Babu PK** for valuable guidance at every stage of this project work.

I am also thankful to project guide **Miss Himna K** for extending their sincere and guidance throughout this work, without their super vision and many hours of devoted guidance, simulating and constructive criticism, this would never come out in this form.

Last but not least, We would like to express my deep and earnest thanks giving to my parents for their moral support and heartfelt cooperation in doing the project. I would also like to thank my friends whose direct or indirect help has enabled me to complete this work successfully.

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Abstract

The **Smart EduConnect System** is an intelligent educational platform designed to bridge the gap between students, colleges, and administrators through data-driven insights and automation. It enables smooth communication, efficient information management, and smart decision-making in the education sector. The **Admin** oversees the system by verifying colleges, managing departments, courses, scholarships, and job postings while handling user feedback and complaints. **Colleges** can register and manage their academic datasets, admission records, and results while requesting new courses and tracking approval statuses. **Users (students)** can explore educational opportunities, attend AI-powered chatbot-based tests, and receive personalized course and job recommendations based on their performance and interests.

A key highlight of Smart Edu Connect is its use of **AI for admission prediction and personalized recommendations**. By analyzing user data, academic history, and college statistics, the system predicts a student's chance of admission to specific institutions and suggests suitable courses or jobs accordingly. This helps students make informed academic and career decisions while assisting colleges in attracting the right candidates. With features like notifications, complaint handling, and smart analytics, Smart EduConnect transforms traditional educational processes into a connected, intelligent, and user-centric digital ecosystem

1. Introduction

The **Smart EduConnect System** is an intelligent, AI-driven educational platform designed to bridge the gap between students, colleges, and administrators through data-driven insights and automation. The system aims to modernize and simplify educational management by enabling smooth communication, efficient information handling, and smart decision-making across the education sector.

In this system, the **Admin** plays a central role by verifying colleges, managing departments, courses, scholarships, and job postings, and handling user feedback and complaints. **Colleges** can register on the platform to manage academic data, admission records, and student results, as well as request new courses and track their approval status. **Students (users)** can explore educational opportunities, participate in AI-powered chatbot-based assessments, and receive personalized course and job recommendations based on their interests and performance.

A key highlight of the Smart EduConnect System is the integration of **Artificial Intelligence** for admission prediction and personalized recommendations. By analyzing student profiles, academic history, and institutional data, the system predicts the likelihood of admission to specific colleges and suggests suitable courses or career options. This empowers students to make informed academic and career decisions while helping colleges attract eligible and suitable candidates.

With features such as real-time notifications, complaint management, and intelligent analytics, **Smart EduConnect** transforms traditional educational processes into a connected, intelligent, and user-centric digital ecosystem

2. System Study & Analysis

2.1 Preliminary Investigation

The preliminary investigation focuses on understanding the existing educational management system and identifying its limitations, challenges, and opportunities for improvement. In the current scenario, students, colleges, and administrators operate on fragmented platforms with limited integration, resulting in inefficient communication, manual processes, and poor decision-making support. Students often struggle to find accurate information about colleges, courses, admissions, scholarships, and job opportunities, while colleges face difficulties in managing academic data and reaching suitable candidates effectively.

To address these challenges, the **Smart EduConnect System** is proposed as an intelligent, AI-driven educational platform that bridges the gap between students, colleges, and administrators through automation and data-driven insights. The system centralizes educational information and enables smooth interaction among all stakeholders. The Admin acts as the core authority, responsible for verifying colleges, managing departments, courses, scholarships, job postings, and handling user complaints and feedback to ensure system reliability and transparency.

Colleges registered on the platform can efficiently manage admissions, academic records, student results, and course requests while tracking approval status in real time. Students can explore educational opportunities, participate in AI-powered chatbot-based assessments, and receive personalized recommendations for courses and careers based on their interests, academic performance, and institutional data.

A major outcome of the preliminary investigation is the identification of Artificial Intelligence as a key enabler for improving educational decision-making. By incorporating AI-based admission prediction and recommendation mechanisms, the proposed system helps students make informed academic and career choices while assisting colleges in identifying suitable applicants. Overall, the Smart EduConnect System aims to transform traditional educational processes into a connected, intelligent, and user-centric digital ecosystem.

2.2 Existing System

The existing education management systems are largely **fragmented, manual, and non-intelligent**, making them inefficient for modern educational needs. Information related to colleges, courses, admissions, scholarships, and job opportunities is scattered across multiple platforms, websites, and offline sources. As a result, students are forced to spend significant time searching and comparing information, which often leads to **confusion, misinformation, and poor decision-making**.

Most current systems lack **data-driven analysis** and rely heavily on manual processes or assumptions. Admission predictions are generally based on guesswork rather than historical data or student performance metrics, reducing accuracy and reliability. There is no mechanism to analyze a student's academic background, interests, or skills to provide **personalized recommendations** for colleges, courses, or career paths.

Additionally, communication between students, colleges, and administrators is highly limited. Important updates regarding admissions, course availability, scholarships, and placement opportunities are often delayed or missed entirely due to the absence of a centralized communication platform. Colleges also face challenges in managing large volumes of student data efficiently, leading to duplication of work and increased administrative burden.

The lack of **automation and system integration** further increases operational costs and the likelihood of human errors. Existing systems do not provide real-time insights, performance analytics, or predictive guidance, which are essential for informed decision-making. Overall, the current education management environment fails to leverage modern technologies such as data analytics and intelligent automation, highlighting the urgent need for a **smart, integrated, and intelligent educational management system**.

2.3 Proposed System

The proposed **Smart EduConnect System** is an **AI-powered, centralized, and data-integrated educational management platform** designed to overcome the limitations of traditional education systems. The system transforms the educational process into a **smart, interactive, and student-centric experience** by seamlessly connecting students, colleges, and administrators on a single digital platform.

For students, the system offers **personalized course, college, and job recommendations** based on academic performance, interests, skills, and historical admission data. An **AI-based admission chance prediction module** analyzes previous cut-off trends and student profiles to provide accurate and reliable admission probability insights, enabling students to make confident academic decisions. The integrated **AI chatbot** assists students with entrance test preparation, instant query resolution, guidance on courses and careers, and system navigation, ensuring continuous support.

Colleges benefit from an **automated and efficient management system** that simplifies admission processes, course management, result publication, and student data handling. The platform reduces manual workload, minimizes errors, and ensures real-time data availability. Colleges can also post job opportunities, scholarships, and announcements directly to students, improving engagement and placement outcomes.

The Admin module plays a crucial role in maintaining system integrity by **verifying colleges**, managing departments, courses, scholarships, and job postings, and monitoring overall system activities. The admin ensures data authenticity, handles feedback and complaints, and maintains smooth communication between all stakeholders.

The system also includes **real-time notifications and alerts** for admissions, deadlines, results, and opportunities, ensuring that users never miss critical updates. Advanced **data analytics and reporting tools** provide valuable insights into student performance trends, admission patterns, and institutional metrics, supporting informed decision-making at every level.

Overall, the Smart EduConnect System creates a **unified digital ecosystem** that enhances transparency, accuracy, accessibility, and efficiency in education management. By integrating AI, automation, and intelligent analytics, the proposed system delivers a future-ready solution that supports smart decision- making and improves the overall quality of the educational experience.

2.4 Feasibility Study

A feasibility study is a preliminary analysis conducted to evaluate and document the viability of the proposed Smart Edu Connect System. The findings of this study help in making an informed decision on whether to proceed with the project. If approved, the feasibility study serves as a foundation to predict the likelihood of success before full-scale system development begins.

This study explores possible solutions to the problems faced in traditional education management systems and recommends the most effective alternative. It evaluates whether an AI-driven educational platform that integrates students, colleges, and administrators is more efficient and reliable than existing manual or semi-automated systems.

The feasibility study identifies the key characteristics of the Smart EduConnect System and gathers detailed information about the required technologies, resources, and implementation strategies from various sources. It also specifies system inputs such as student profiles, academic records, and college data, along with expected outputs like admission predictions, personalized recommendations, and notifications.

A central component of this study is the cost–benefit analysis, which compares the costs of implementing the system against the benefits it delivers. While some benefits are measurable in financial terms, others are qualitative, including improved transparency, better decision-making, reduced administrative workload, and enhanced student satisfaction.

A feasibility study is necessary for the development of the Smart EduConnect System for the following reasons:

- Existing education systems are fragmented, manual, and inefficient.
- Students lack accurate guidance for course selection and admission chances.
- Advancements in AI and data analytics enable intelligent recommendations.
- Institutions require better communication and centralized management.
- Competing platforms are increasingly adopting AI-driven educational solutions.

The feasibility study determines whether the proposed system can be developed using available resources and whether the expected benefits justify the investment

2.4.1 Technical Feasibility

The Technical Feasibility of the Smart EduConnect System examines whether the required technology is available and whether the system can be developed with existing technical expertise. This evaluation is based on a preliminary system design that defines system inputs, outputs, databases, modules, and workflows.

The system will be developed using standard and widely supported technologies such as Python, Django, MySQL, and web-based interfaces. AI and machine learning techniques will be used for admission prediction and personalized recommendations by analyzing student performance data and college statistics.

The system requires standard computing infrastructure and internet connectivity. Since all required software tools and development environments are readily available, the project does not face any major technical limitations. Hence, the system is technically feasible

2.4.2 Operational Feasibility

Operational Feasibility evaluates how effectively the Smart EduConnect System will function once implemented. The system is designed to be user-friendly and role-based, allowing administrators, colleges, and students to interact with the platform efficiently.

Training and guidance will be provided to users to ensure smooth adoption. Automated processes such as notifications, complaint handling, and recommendation generation reduce manual effort and improve operational efficiency.

The proposed system offers the following operational advantages:

- Easy-to-use and intuitive interface
- Faster access to accurate academic and admission-related information
- Improved communication between students, colleges, and administrators
- Reduced workload for educational institutions

Due to these benefits, the Smart EduConnect System is operationally feasible and expected to gain wide acceptance.

2.4.3 Economic Feasibility

Economic Feasibility examines whether the organization can afford to develop and maintain the Smart EduConnect System and whether the benefits outweigh the costs involved. The initial investment includes software development, system hosting, and maintenance costs.

However, the system significantly reduces administrative expenses by automating admission guidance, communication, and data management. The use of open-source tools further minimizes development costs.

The economic feasibility focuses on:

- Reduction in manual processing and paperwork
- Improved efficiency and decision-making accuracy
- Long-term cost savings for institutions
- High return on investment through better system utilization

Thus, despite initial costs, the long-term benefits justify the investment, making the system economically feasible..

2.4.4 Behavioral Feasibility

Behavioral Feasibility analyzes how users respond to the introduction of the new system. Since Smart Edu Connect simplifies complex educational processes and provides clear benefits, user resistance is expected to be minimal.

Students benefit from personalized guidance and predictions, colleges gain better applicant

management, and administrators achieve centralized control. Awareness programs and training sessions will further support user acceptance.

Therefore, the system is behaviorally feasible and likely to be positively received.

2.4.5 Legal Feasibility

Legal Feasibility determines whether the Smart EduConnect System complies with applicable laws and regulations. Since the system handles sensitive student and institutional data, it must adhere to data protection and privacy regulations.

Appropriate authentication, authorization, and data security measures will be implemented to prevent misuse of information. As long as privacy policies and ethical data-handling practices are followed, the system does not violate any legal requirements.

Hence, the Smart Edu Connect System is legally feasible

2.4.6 Schedule Feasibility

The Schedule Feasibility of the Smart Edu Connect System evaluates whether the project can be completed within a reasonable timeframe and whether the proposed development timeline is practical. Given the modular structure of the system, development can be carried out in well-defined phases.

The implementation timeline is divided as follows:

- Phase 1: Requirement analysis and system design
- Phase 2: Development of core modules (Admin, College, Student)
- Phase 3: Integration of AI-based admission prediction and recommendation system
- Phase 4: Testing, debugging, and performance optimization
- Phase 5: Deployment and user training

With proper planning, resource allocation, and use of existing development frameworks, the project can be completed within the estimated academic schedule. Therefore, the Smart EduConnect System is schedule-feasible

The implementation timeline will be structured in phases, including:

- **Phase 1**: System design and hardware setup.
- **Phase 2**: Development and testing of AI-based facial recognition
- **Phase 3**: Pilot testing within a small group of students.
- **Phase 4**: Full-scale implementation and user training.

2.5 Project Planning and Scheduling

2.5.1. Project Planning

For the successful completion of the Smart EduConnect System project, detailed planning and scheduling are essential. The software development process is carried out in multiple phases to ensure systematic execution and timely completion of the project.

The first step is the Requirement Analysis phase. During this phase, extensive research is conducted by studying existing education management systems, interacting with students, college administrators, and faculty members, and analyzing current admission guidance and communication methods. Discussions with project guides and domain experts help in understanding system expectations and identifying key challenges faced by users.

After gathering the required information, a Preliminary Investigation is carried out by addressing the following key questions:

1. How are students currently searching for colleges, courses, and job opportunities?
2. How do colleges manage admissions, courses, and academic data?
3. What limitations exist in the current education management systems?
4. How serious are the issues related to manual searching and lack of guidance?
5. How frequently do students and institutions face these problems?
6. What are the main causes behind inefficient communication and decision-making?

based on the findings of the analysis phase, the system requirements are clearly identified and divided into different functional modules such as Admin, College, Student, AI-based Recommendation, Admission Prediction, and Chatbot modules.

The System Design phase follows, where a structured blueprint of the Smart EduConnect System is created. This phase includes the preparation of Data Flow Diagrams (DFDs), Entity Relationship (ER) diagrams, database schema, and overall system architecture. The design phase ensures proper data flow, system scalability, and role-based access control.

The next phase is System Development, where the approved design is converted into executable code. The system is developed using Python and Django for backend processing, MySQL for database management, and web technologies such as HTML, CSS, and JavaScript for the user interface. AI and machine learning techniques are integrated for admission prediction and personalized recommendations, while a chatbot module is implemented for testing and guidance. Proper documentation is maintained throughout this phase to support future maintenance and upgrades.

After development, the system enters the Testing Phase, where functionality, performance, and security testing are conducted. Each module is tested individually and collectively to ensure that the system meets user requirements and operates without errors.

The Implementation Phase involves deploying the Smart EduConnect System in a real-time environment. During this phase, administrators, college representatives, and students are introduced to the system. Necessary training and demonstrations are provided to ensure smooth operation and effective usage.

Like any software system, the Smart EduConnect System requires Maintenance and Support after deployment. Maintenance activities include monitoring system performance, fixing bugs, updating features, and improving system security. Regular updates ensure that the system remains efficient, reliable, and adaptable to future educational requirements.

3. System Requirement Specification

3.1.Introduction

A System Requirements Specification (SRS) is a comprehensive description of the Smart Edu Connect System to be developed, outlining both functional and non-functional requirements. Functional requirements describe the core features and operations of the system, while non-functional requirements define constraints related to performance, security, usability, and reliability

The SRS document provides a clear understanding of how students, colleges, and administrators interact with the system. It may include use cases that illustrate user interactions such as college verification, admission prediction, course recommendation, and complaint handling. This document ensures that all essential system requirements are clearly defined and documented before development begins.

To derive accurate requirements, a thorough understanding of the proposed system is required. This is achieved through detailed research, analysis of existing education management systems, and continuous discussions with stakeholders such as students, college administrators, and project guides. The requirements are refined through iterative evaluation until the system is successfully implemented.

3.2 Purpose

The purpose of the Smart EduConnect System project is to overcome the limitations of traditional education management systems by providing an intelligent, centralized, and data-driven platform that supports informed academic and career decision-making.

The project aims to achieve the following objectives:

* Improve Educational Guidance – Provide AI-based admission predictions and personalized course and job recommendations.

* Enhance Communication – Enable seamless interaction between students, colleges, and administrators through a unified platform.

* Automate Academic Management – Simplify college verification, course management, and admission record handling.

* support Student Decision-Making – Help students choose suitable colleges and career paths based on performance and interests.

* Increase Transparency – Provide real-time notifications, approval tracking, and complaint handling.

* Optimize Administrative Efficiency – Reduce manual workload through automation and centralized data management.

Overall, the purpose of the Smart EduConnect System is to transform traditional educational processes into an intelligent, connected, and user-centric digital ecosystem..

3.3. Scope

The scope of the Smart EduConnect System includes the design, development, implementation, and operation of an AI-powered educational platform. The system integrates academic data management, admission prediction, recommendation engines, chatbot-based testing, and communication services.

The system focuses on enabling students to explore colleges, courses, scholarships, and jobs while receiving intelligent guidance. Colleges can manage admissions, academic datasets, and course requests, while administrators oversee verification, monitoring, and system control.

By clearly defining the scope, the project ensures effective resource allocation, clear objectives, and successful delivery of a solution that improves accessibility, accuracy, and efficiency in education management educational institutions.

3.4. System Requirements

3.4.1 Hardware Specification

The selection of hardware plays an important role in the smooth functioning of the **Smart Edu Connect System**. The system requires standard computing resources to ensure reliable performance and efficient processing of data.

A **64-bit Intel Pentium Core i3 processor or above** is required to support system operations and multitasking. A minimum of **4GB RAM** ensures smooth execution of applications without delays. The system needs **512GB or higher hard disk space** to store user data, academic records, and system files securely.

A **stable internet connection** is essential for real-time data access, updates, notifications, and seamless communication between students, colleges, and administrators.

- Processor : Intel Pentium Core i3 and above, 64 bits
- RAM:Min4GB RAM
- HARD DISK:512GB or above
- Internet Connection:Required for real-time access

3.4.2 Software Specification

Selecting appropriate software is essential for meeting system requirements and ensuring efficient performance of the **Smart EduConnect System**. The system is developed using reliable and widely used technologies to provide stability and scalability.

The application runs on **Windows 8 or above**. The **front end** is developed using HTML, CSS, Bootstrap, and JavaScript for a responsive web interface, while **Flutter** is used to build cross-platform mobile applications. The **back end** is implemented using **Python Django** with **Dart** supporting Flutter- based interactions. **MySQL** is used as the database to securely store and manage system data.

Development tools such as **JetBrains PyCharm** and **Android Studio** are used for efficient coding, testing, and debugging. The system uses **Django technology and framework**, which supports secure development, modular design, and easy maintenance.

- OPERATING SYSTEM: WINDOWS 8 or above
- FRONT END: HTML, CSS,Bootstrap,Javascript,Flutter
- BACK END: Python Django - flutter Dart
- DATABASE:Mysql
- IDE USED: JetbrainsPycharm, Android studio
- TECHNOLOGY USED: DJANGO
- FRAME WORK USED: DJANGO

4. System Design

The most creative and challenging phase of the system life cycle is system design. The term design describes both the final system and the process by which it is developed. It refers to the technical specifications that will be applied in implementing the Smart Edu Connect System (SECS). This phase includes the construction of the program, system architecture, database design, and program testing. The key question addressed during this stage is: “How should the problem be solved?”

System design provides a structured solution for determining the approach to developing a new system. This important phase consists of multiple steps that define the procedural and technical details required for system implementation based on the results of the feasibility study. The primary focus is on converting functional and performance requirements into detailed design specifications.

The design process of the Smart Edu Connect System includes both logical and physical system design. Logical design defines the system workflow, data flow, user interactions, and functional modules such as student management, college management, AI-based admission prediction, recommendation engine, chatbot testing, notifications, and complaint handling. Physical design focuses on the actual implementation environment, including hardware requirements, software platforms, database structures, and network configurations.

Input and output design plays a crucial role in system design. The first step involves identifying how outputs such as admission predictions, course recommendations, job suggestions, reports, and notifications will be generated and presented to users. Sample input forms and output screens are designed to visualize system behavior and ensure user-friendly interaction. Based on output requirements, input data formats and master files are designed to store student profiles, academic records, college details, courses, and job information efficiently.

The operational design phase includes program construction, module-wise testing, and system integration. All required programs and modules are listed and developed according to the design specifications. Comprehensive testing strategies are applied to verify system functionality, data accuracy, security, and performance. Proper system documentation is prepared to support future maintenance and upgrades.

Finally, the system design phase evaluates the justification of the Smart Edu Connect System and its impact on students, colleges, and administrators. A final system design report is prepared before implementation, which includes process flow diagrams, data flow diagrams, database structures, input/output layouts, and a detailed implementation plan to ensure successful deployment of the system

4.1. Module Description

- **Admin**
- **College**
- **User**

- ADMIN
  - Login
  - View users
  - Verify College
  - Manage Department
  - Manage Course
  - Manage Article
  - Send notification
  - Manage Scholarship
  - Manage Jobs
  - View complaints & Send Reply

- COLLEGE
  - Registration
  - Login
  - Manage dataset
  - Manage Last Admission details
  - Manage Result
  - Request For Course
  - Update Course fee
  - View Request Status
  - View Notification
  - Update Profile

- USER
  - Register
  - Login
  - Attend Test
  - Suggest College
  - View College,Course and Fees
  - View Scholarship
  - Course Recommendation
  - Job Recommendation
  - Send complaints & view reply
  - Admission Chance Prediction
  - Change Password
  - Update Profile

4.2. Data Flow Diagram(DFD)

A data flow diagram (DFD) or a bubble chart is a graphical tool for structured analysis. DFD models a system by using external entities from which dataflow to a process, which transforms the data and creates output data flows which go other process or external entities or files. Data in files may also flow to processes as inputs.

DFDs can be hierarchically organized, which help in partitioning and analyzing large systems. As a first step, one data-flow diagram can depict an entire system which gives the system overview. It is called context diagram of level0 DFD. The context diagram can be further expanded.

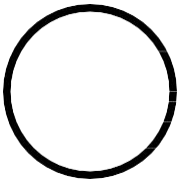
The successive expansion of a DFD from the context diagram to those giving more details is known as leveling of DFD. Thus a top down approach is used starting with an overview and then working out the details. The main merit of the DFD is that it can provide an overview of what data a system would process, what transformation of data are done, what files are used, and where the results flow.

DFD Design Notation

In DFD, there are four main symbols:



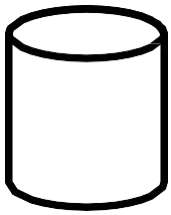
Source of Destination of Data



Process

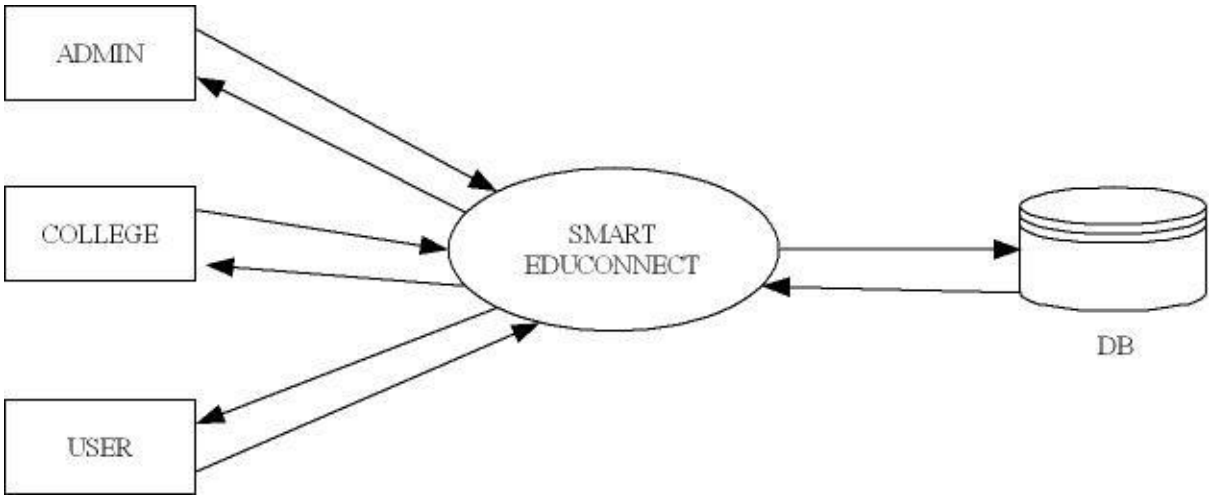


Flow of Data

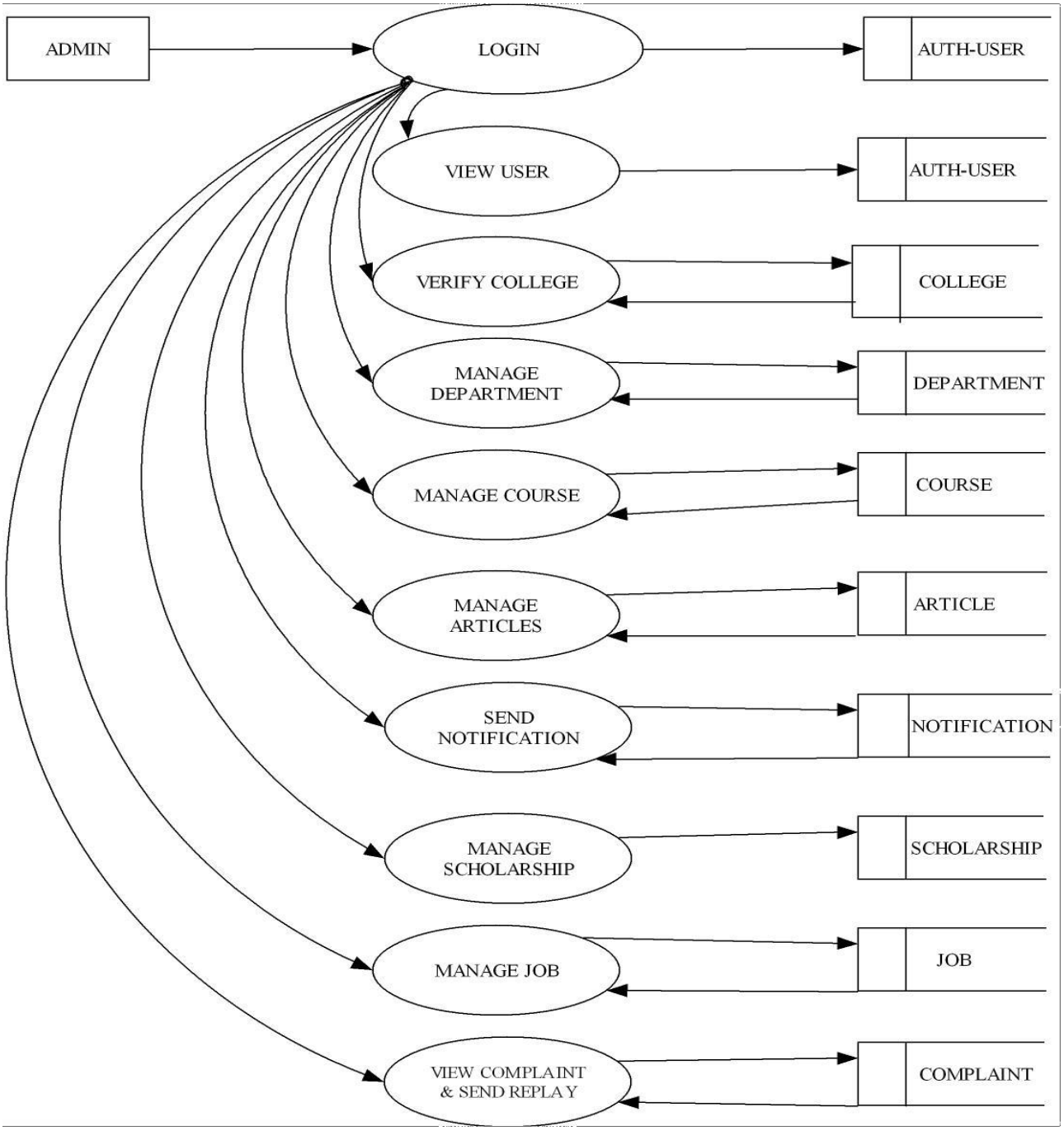


Repository of Data

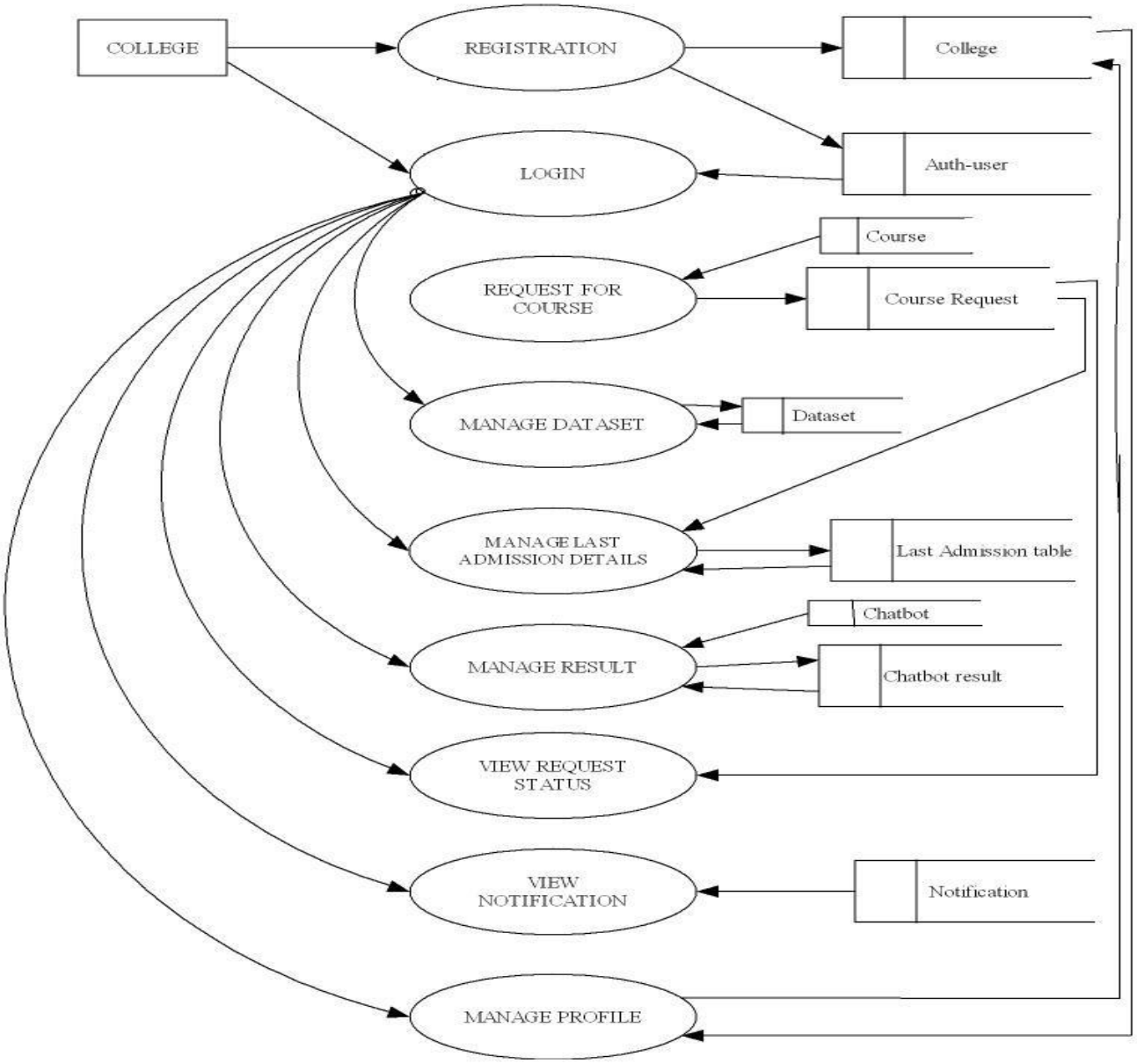
Level 0 DFD



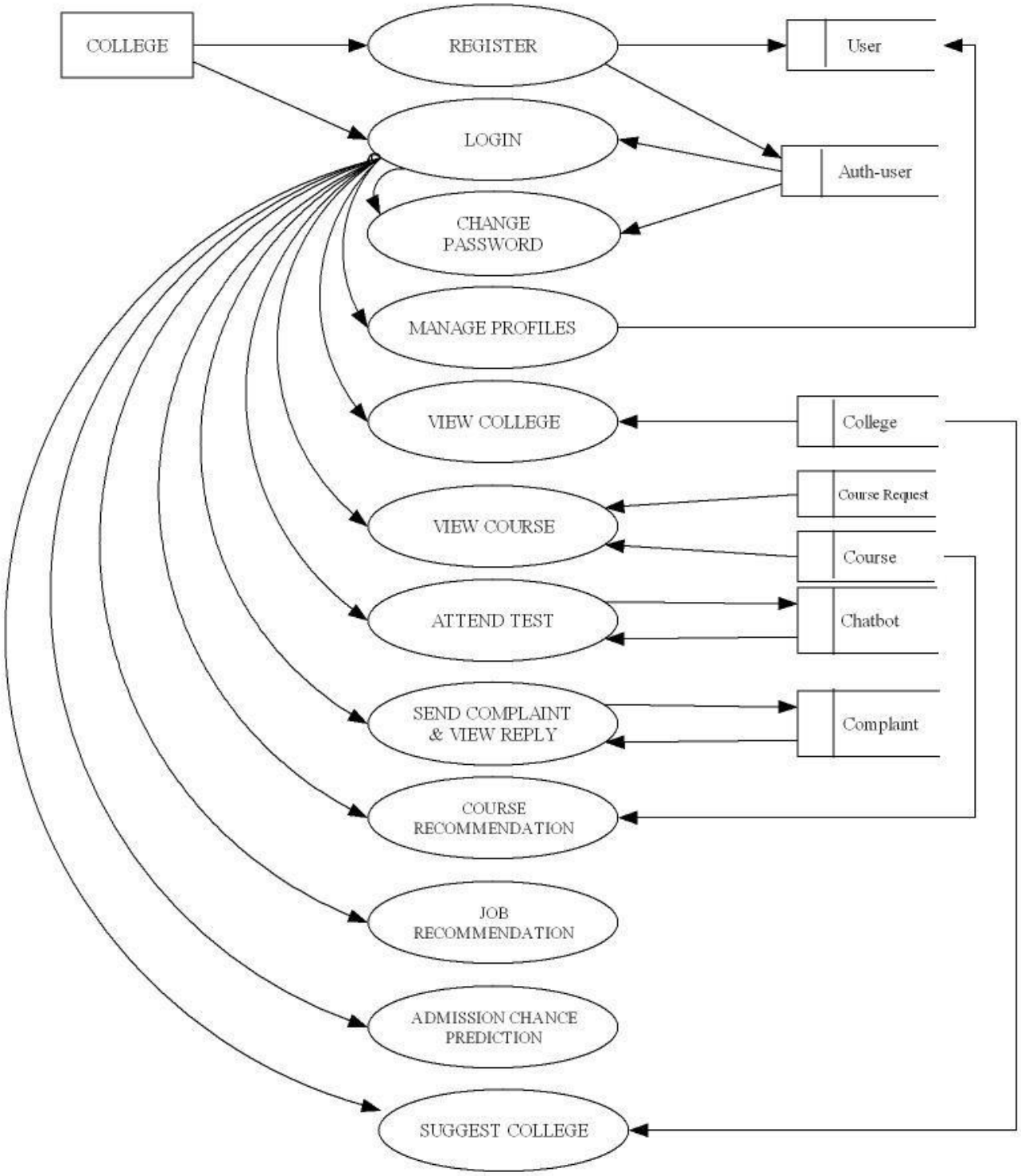
Level 1.1 DFD


ADMIN
LOGIN
AUTH-USER
VIEW USER
AUTH-USER
VERIFY COLLEGE
COLLEGE
MANAGE DEPARTMENT
DEPARTMENT
MANAGE COURSE
COURSE
MANAGE ARTICLES
ARTICLE
SEND NOTIFICATION
NOTIFICATION
MANAGE SCHOLARSHIP
SCHOLARSHIP
MANAGE JOB
JOB
VIEW COMPLAINT & SEND REPLAY
COMPLAINT

Level 1.2 DFD


COLLEGE
REGISTRATION
College
LOGIN
Auth-user
Course
REQUEST FOR COURSE
Course Request
MANAGE DATASET
Dataset
MANAGE LAST ADMISSION DETAILS
Last Admission table
Chatbot
MANAGE RESULT
Chatbot result
VIEW REQUEST STATUS
VIEW NOTIFICATION
Notification
MANAGE PROFILE

Level 1.3 DFD


COLLEGE
REGISTER
User
LOGIN
Auth-user
CHANGE PASSWORD
MANAGE PROFILES
VIEW COLLEGE
College
Course Request
VIEW COURSE
Course
ATTEND TEST
Chatbot
SEND COMPLAINT & VIEW REPLY
Complaint
COURSE RECOMMENDATION
JOB RECOMMENDATION
ADMISSION CHANCE PREDICTION
SUGGEST COLLEGE

4.3. Table Design

User

Field Name	Data Type	Constrains
Id	Int	Primary key
Login	Varchar(25)	Foreignkey
Name	Varchar (200)	Not null
Gender	Varchar (200)	Not null
Place	Varchar (200)	Not null
Post	Varchar (200)	Not null
Pin	Integer	Not null
Phone	BigInt	Not null
Email	Varchar (200)	Not null
Image	file	null
Proof	file	null

Data set

Field Name	Data Type	Constraints
Id	Int	Primary key
Course_Req	Integer	Foreign key
Question	Varchar (200)	Not null
Answer	Varchar (200)	Not null

College

Field Name	Data Type	Constraints
Id	Int	Primary key
Login	Integer	Foreignkey
Name	Varchar(200)	Not null
Place	Varchar(200)	Not null
Email	Varchar(200)	Not null
Phone	BigInt	Not null
Status	Varchar(50)	Default=”pending”
proof	file	null

Notification

Field Name	Data Type	Constrains
Id	Int	Primary key
Notification	Varchar(200)	Not null
details	Varchar(200)	Not null
Date	Date	Not null

Chatbot_result

Field Name	Data Type	Constraints
Id	Int	Primary key
CHATBOT	Varchar	Foreign key
Result	Varchar (200)	Not null

Course

Field Name	Data Type	Constraints
Id	Int	Primary key
Department_id	Int	Foreign key
course	Varchar (200)	Not null
Details	Text	Not null
Fee	Flout	Not null

Course_Request

Field Name	Data Type	Constraints
Id	Int	Primary key
COLLEGE	Varchar	Foreign key
COURSE	Varchar	Foreign key
Date	Date	Not null
Status	Varchar (200)	Not null

Department

Field Name	Data Type	Constraints
Id	Int	Primary key
Department	Varchar (200)	Not null
Details	Varchar (200)	Not null

Complaint

Field Name	Data Type	Constraints
Id	Integer	Primary key
USER_ID	Integer	Foreign key
Complaint	Varchar (200)	Not null
Date	Date	Not null
Reply	Varchar (020)	Not null

Article

Field Name	Data Type	Constraints
Id	Int	Primary key
Article_title	Varchar (200)	Not null
Article_subject	Varchar (200)	Not null
Date	Date	Not null

Chatbot

Field Name	Data Type	Constraints
Id	Int	Primary key
User	Integer	Foreign key
Question	Text	Not null
Answer	Text	Not null
Date	Date	Not null

Last_admission

Field Name	Data Type	Constraints
Id	Int	Primary key
COLLEGE	Varchar	Foreign key
COURSE	Varchar	Foreign key
Year	Integer	Not null
Mark_starting	Flout	Not null
Mark_ending	Flout	Not null
Date	Date	Not null

Suggestion

Field Name	Data Type	Constraints
Id	Int	Primary key
STUDENT_id	Integer	Foreign key
Suggestion	Varchar (200)	Not null
Details	Varchar (300)	Not null
Date	Date	Not null

Change password connection

Field Name	Data Type	Constraints
Id	Int	Primary key
User_id	Int	Foreign key
password	Varchar	Not null

Scholership

Field Name	Data Type	Constraints
Id	Int	Primary key
Name	Varchar (200)	Not null
Details	Varchar (200)	Not null
Date	Date	Not null
Documents	File	null

Job

Field Name	Data Type	Constraints
Id	Int	Primary key
Job_title	Varchar (75)	Not null
Company	Varchar (75)	Not null
Qualification	Varchar (75)	Not null
Start_date	Date	Not null
Last_date	Date	Not null
Position	Varchar (75)	Not null
No_vacancy	Varchar	Not null
Link	Text	Not null

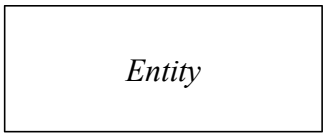
4.4. ER Diagram

An ER diagram can express the overall structure of the database graphically. ER diagrams are simple and clear. ER diagrams often use symbols to represent three different types of information. Boxes are commonly used to represent entities. Diamonds are normally used to represent relationships and ovals are used to represent attributes.

Entity Relationship Diagram Notations

Entity:

An entity is an object or concept about which you want to store information.



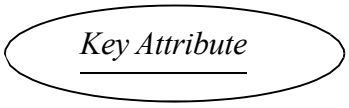
Attribute:

Each entity has attributes, or particular properties that describe the entity.



Key attribute :

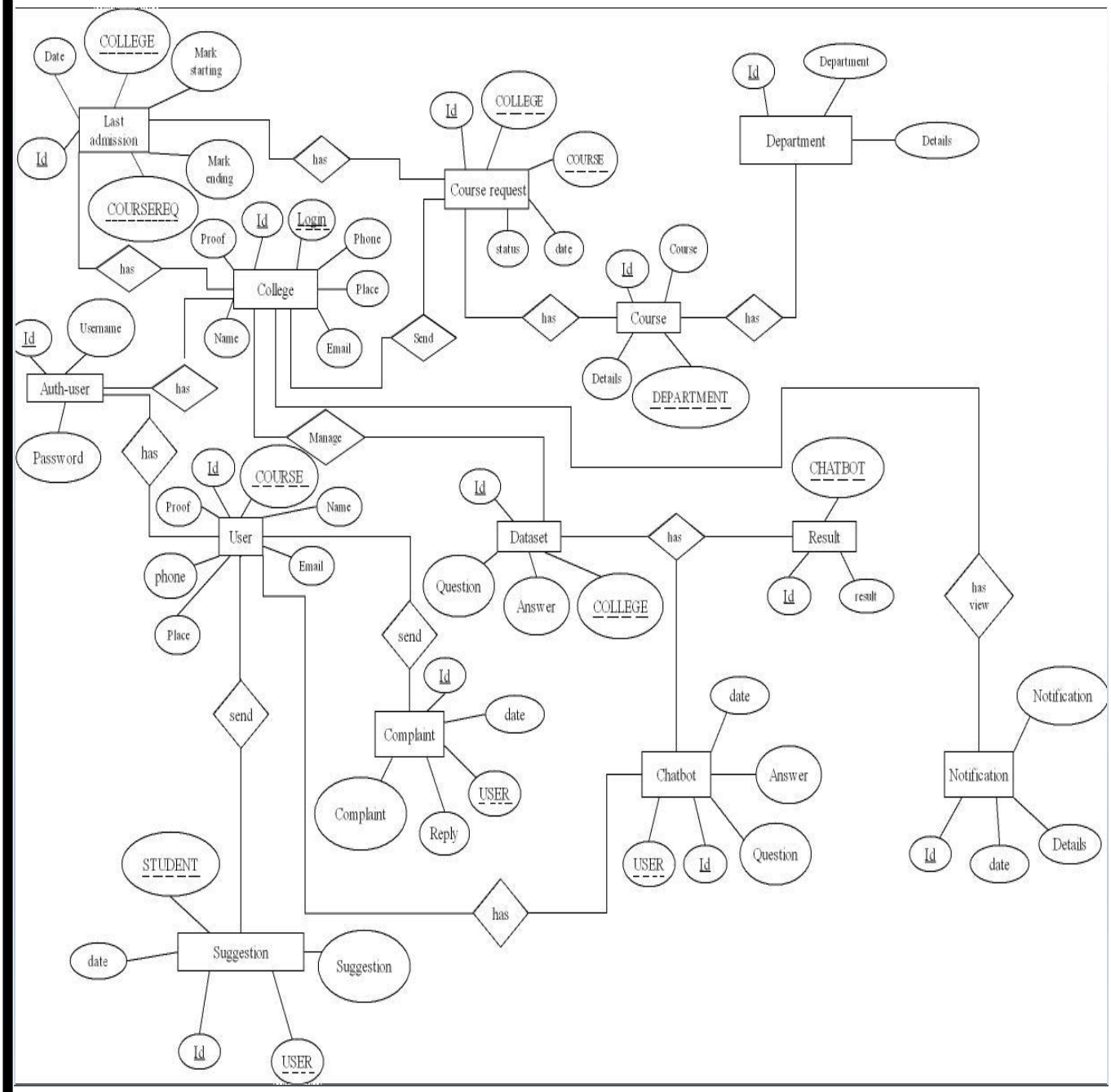
A key attribute is the unique, distinguishing characteristic of the entity.



Relationships:

Relationships illustrate how two entities share information in the database structure.The ER diagram used in this project is given in the next page

ER Diagram


COLLEGE
Date
Mark starting
Last admission
Id
Mark ending
COURSEREQ
has
Id
COLLEGE
COURSE
Course request
status
date
Id
Department
Department
Details
Proof
Id
Login
Phone
Place
College
Name
Email
Send
has
Id
Course
Course
has
Details
DEPARTMENT
Id
Username
Auth-user
has
Password
has
Manage
Id
COURSE
Proof
Name
User
phone
Email
Place
send
send
Id
Complaint
date
Complaint
Reply
USER
Id
Dataset
Question
Answer
COLLEGE
has
CHATBOT
Result
Id
result
has view
date
Chatbot
Answer
USER
Id
Question
Notification
Notification
Id
date
Details
STUDENT
has
date
Suggestion
Suggestion
Id
USER

5. Coding and Implementation

5.1. Coding Environment

5.1.1 Front end

An Integrated Development Environment (IDE) (also known as Integrated Design Environment or Integrated Debugging Environment) is a software application that provides comprehensive facilities to computer programmers for software development. An IDE normally consists of:

- A source code editor
- A compiler and/or an interpreter
- Build automation tools
- A debugger

5.1.2 Android Studio

Android Studio is the official integrated development environment (IDE) for Android app development, developed by Google. Android Studio continues to evolve with updates and new features to support the latest advancements in Android development, making it the preferred IDE for Android app developers worldwide.

Built on the foundation of IntelliJ IDEA, Android Studio provides powerful features such as a Gradle -based build system for efficient project management, an intuitive layout editor for designing UIs, and a versatile code editor with support for Java, Kotlin, and XML. Its built-in Android Emulator enables developers to test apps across various virtual devices seamlessly. With debugging tools, performance profilers, and version control integration, Android Studio facilitates the creation, testing, and optimization of Android apps, making it the preferred choice for developers worldwide.

Here are some key details of android studio:

1. Features: Android Studio offers a comprehensive set of features including code editing, debugging, testing, performance profiling, and a built-in emulator. It supports both Java and Kotlin programming languages for Android app development.

2. User Interface: It has a user-friendly interface with various panels for viewing code, layouts, project structure, and more. It also includes tools for designing user interfaces using XML or a visual layout editor.

3. Gradle Build System: Android Studio uses the Gradle build system, which allows for flexible project configuration and dependency management.

4. Emulator: Android Studio comes with a powerful emulator that allows developers to test their apps on virtual Android devices with different configurations and API levels.

5. Plugins and Extensions: Developers can extend the functionality of Android Studio through plugins and extensions available in the IntelliJ Plugin Repository.

6. Version Control Integration: Android Studio has built-in support for version control systems like Git, making it easy for developers to collaborate on projects and manage code changes.

7. Updates and Support: Google regularly updates Android Studio with new features, improvements, and support for the latest Android SDK versions and APIs.

8. Documentation and Community: Android Studio is supported by extensive documentation,

9. tutorials, and a vibrant community of developers who contribute to forums, Q&A sites, and online resources. Overall, Android Studio provides a robust and efficient environment for building high-quality Android apps, whether for personal projects or commercial applications.

5.1.3 Pycharm

PyCharm is a popular Integrated Development Environment (IDE) specifically designed for Python development. Developed by JetBrains, PyCharm provides a comprehensive set of tools and features to enhance productivity and streamline the development process for Python programmers. Some key features of PyCharm include:

1. Code Editor: PyCharm offers a powerful code editor with features like syntax highlighting, code completion, code refactoring, and smart code navigation.

2. Intelligent Code Assistance: It provides intelligent code assistance, including code inspections, quick fixes, and context-aware code suggestions, to help developers write clean and error-free code.

3. Debugging Tools: PyCharm includes built-in debugging tools that allow developers to debug Python code easily, set breakpoints, inspect variables, and analyze program execution flow.

4. Testing Support: PyCharm supports various testing frameworks for Python, such as unittest, pytest, and doctest, enabling developers to write and run tests directly within the IDE.

5. Version Control Integration: It seamlessly integrates with version control systems like Git, Mercurial, and Subversion, allowing developers to manage code changes, commit revisions, and collaborate with team members effectively.

6. Virtual Environment Management: PyCharm simplifies the management of Python virtual environments, enabling developers to create, activate, and configure virtual environments for their projects effortlessly.

5.1.4 Backend

The backend of a software application refers to the server-side components responsible for processing data, managing user requests, and performing computations. It includes databases, application logic, server, and APIs that handle communication between the server and the client- side interface. In essence it's the backbone that supports the functionality and performance of an application, hidden from the end user's view.

5.1.5 Python

Python is a high-level, interpreted programming language known for its simplicity and readability. It emphasizes code readability and productivity, making it popular among beginners and experienced programmers alike. Python supports multiple programming paradigms, including procedural, object-oriented, and functional programming. It has a vast standard library and a vibrant ecosystem of third-party packages, making it suitable for a wide range of applications, including web development, data analysis, artificial intelligence, machine learning, automation, and scientific computing.

5.1.6 Datastore

A Datastore is a centralized repository that stores, manages, and organizes data. It can come in various forms, including databases, data warehouses, and data lakes. Datastores are essential components of software systems,. They can handle various types of data, such as structured, semi-structured, and unstructured data, and support different data models, including relational, MySQL, and graph databases. Datastores play a crucial role in applications ranging from small- scale web applications to large-scale enterprise systems, scalability, and reliability.

5.1.7 MySQL

MySQL is an open-source relational database management system (RDBMS) that is widely used for managing structured data. It is developed, distributed, and supported by Oracle Corporation. Here are some key details about MySQL:

1. Relational Database: MySQL follows the relational model and uses Structured Query Language (SQL) for managing and manipulating data.

2. Open Source: MySQL is open-source software, which means it is freely available for use, modification, and distribution under the terms of the GNU General Public License (GPL). There is also a commercial version available with additional features and support from Oracle Corporation.

3. Cross-Platform Compatibility: MySQL is available for various operating systems, including Windows, macOS, Linux, and Unix, making it suitable for a wide range of applications and environments.

4. Scalability: MySQL is designed to scale from small, single-user applications to large, enterprise-grade systems with millions of users and terabytes of data. It supports features like replication, clustering, and partitioning to handle high-volume workloads and ensure high availability and performance.

5. High Performance: MySQL is known for its fast performance and efficiency. It uses various optimization techniques, such as indexing, caching, and query optimization, to deliver fast response times and throughput.

6. Security: MySQL provides robust security features to protect data integrity and confidentiality. It supports features like authentication, access control, encryption, and auditing to ensure secure access and data protection.

7. dframeworks, and tools, making it a popular choice for building web applications, e-commerce platforms, content management systems (CMS), and other types of software solutions.

Overall, MySQL is a powerful and versatile database management system that offers reliability, performance, scalability, and security for managing structured data in a wide range of applications and environments.

Programming Languages

Dart

Dart is the programming language used in Flutter for building cross-platform mobile, web, and desktop applications. It is an object-oriented, class-based language developed by Google and is known for its simplicity, speed, and flexibility. Dart was specifically designed with Flutter in mind, providing a seamless and efficient development experience for creating high-performance applications. Dart offers developers a familiar and structured programming paradigm, featuring a strong yet optional sound type system. This system allows for static analysis, ensuring robust type safety during development while offering flexibility for type inference where needed. Dart's support for both Just-in-Time (JIT) and Ahead-of-Time (AOT) compilation is a pivotal feature, enabling rapid development cycles through JIT and then effortlessly transitioning to AOT for deploying optimized, production- grade code, thereby maintaining high-performance standards.

Framework

Django

Django is a high-level Python web framework that encourages rapid development and clean, pragmatic design. It follows the Model-View-Template (MVT) architectural pattern, emphasizing reusability and "pluggability" of components. Django comes with built-in features for authentication, URL routing, database ORM (Object-Relational Mapping), templating engine, and admin interface, making it ideal for building web applications efficiently. It prioritizes security, scalability, and maintainability, making it a popular choice for developers worldwide.

Flutter

Flutter is an open-source UI software development kit created by Google for building natively compiled applications for mobile, web, and desktop from a single codebase. It uses the Dart programming language and offers a rich set of pre-built widgets for crafting modern and responsive user interfaces. Flutter employs a reactive framework, allowing developers to create flexible layouts and interactive experiences with hot reload for fast iteration. It's known for its high performance, expressive UI, and extensive community support, making it a popular choice for cross-platform app development.

5.2 Implementation

Implementation of the system refers to the final installing of the package in its real environment , to the satisfaction of the indeed users and the operation of the system. It is the process of converting a new or revised system design to operation. It is the key stage in achieving successful new system. The process of putting the developed system in actual use is called system implementation. This includes all those activities that take place to convert from the old system to new system. It must therefore be carefully planned and controlled. Proper guidance should be imparted to the users so that he is comfortable in using the application.

Implementation Plan

The transformation from theoretical designs to working system is done in this stage. Developed package of system is tested with simple data, accurate error identification and then through proposed change from the user etc. a dress rehearsal working of system is done, so as the system is scrutinized, for pointing out errors and modifications required if any keeping in mind the expectations and specifications from the system.

Education And Training

The expectations from the system are achieved by the people who will be involved to be confident of their role in the new system. The complexity of the system is directly proportional to the amount of training and education given for the user .Education is different from the training, as the user through education can be a part of development of the system. Education has the capability to make training more interesting and important contributions in the system changes.

Training just means to give user specific skills in order to meet their new job requirements. The role of system analyst in training will make it more understandable and effective. Training provides a better overview of new system and its present objectives

Security

System security is a branch of technology known as information security as applied to computers and networks. The objective of system security includes protection of information and property from theft, corruption, or natural disaster, while allowing the information and property to remain accessible and productive to its intended users. The term system security, means the collective processes and mechanisms by which sensitive and valuable information and services are protected from publication, tempering or collapse by unauthorized activities or untrustworthy individuals and unplanned events respectively. The technologies of system security are based on logic. As security is not necessarily the primary goal of most computer applications, designing a program with security in mind often imposes restrictions on that program's behavior.

Post Implementation Review

System performance v/s expected requirements are evaluated. The implementation problems if any is taken seriously and taken care of along with admiring the achievements, failures etc. The works done here are used to improve the efficiency and user friendliness of the system.

Maintenance

Maintenance is making adaptation of the software for external changes (requirements changes or enhancements) and internal changes (fixing bugs). When changes are made during the maintenance phase all preceding steps of the model must be revisited.

There are three types of maintenance:

1. Corrective (Fixing bugs/errors)
2. Adaptive (Updates due to environment changes)
3. Perfective (Enhancements, requirements changes

Maintenance is enigma of the system development. The definition of the software maintenance can be given describing four activities that are undertaken after the program is released for use. The maintenance activity occurs since it is unreasonable to assume that software testing will uncover all in a large system. The second activity that contributes the definition of maintenance occurs since rapid changes are encountered in every aspects of computing. The third activity involves recommendation for new capabilities, modification to the existing functions and general enhancements when the software is used. The fourth maintenance activity occurs when software is changed to improve future maintainability or reliability.

6. Testing

6.1. Testing Objectives

Testing is an important step in the software engineering process that could view rather than constructive. Testing is the process of executing a program with the intent of finding an error. a good test is that has probability to find an as yet undiscovered error.

- A good case is one that has a high probability of finding an unpredictable error.
- A successful case is one that has a high probability of finding an unpredictable error.
- A good test case is one that provides solution to that unpredictable error.
- A test plan entailed the following activities. We prepare list plan.
- We specified condition for users acceptance testing.
- We prepared list data for program testing.
- Also we prepared list data transaction plan testing.
- Then we planned user training.
- Our programs were compiled and assembled.
- Job performance aids were prepared.

Need For Testing

Software testing is a critical element of software quality assurance and represents the ultimate review of specification, design and coding . testing includes the verification of the basic logic of each program and verifies that the entire system works properly. Testing the individual program involves and attempt to be sure of the most likely possible. Test case design focuses on asset of technique for the creation test the cases that meet over all testing objectives.

Test Results

Test results emphasize how the actual results differed from the expected results. This suggests the need for re-testing, and to discover the source of differences. The test program or application with the intent of finding software bugs phase of systems development process involves the defining of the criteria by which the system will be tested and measuring the criteria against the acceptable failure rate. Individual modules are tested during the development itself. Errors detected are corrected and re-tested, and the project leader has verified the compliance. Each

input, output and processes are tested to verify that it performs as specified in the design. The units in the system are re-compiled and errors found are corrected as indicated by the compiler. The tests are repeated until all known errors are eliminated and the program matched the design specifications. Separate tests are performed to ensure that program units are properly interfaced with each other to form a complete system.

6.2. Testing Strategy

Software testing determines the correctness, completeness, and quality of software being developed. Validation refers to the process of checking that the developed software meets the requirements specified by the user. The activities involved in the testing phase basically evaluate the capability of that system meets its requirements. The main objective of software testing is to detect errors in the software. Errors occur if some part of the developed system is found to be incorrect, incomplete or inconsistent. Test techniques include, but are not limited to, the process of executing a program or application with the intent of finding software bugs (errors or other defects). It involves the execution of a software component or system to evaluate one or more properties of interest. In general, these properties indicate the extent to which the component or system under test:

- meets the requirements that guided its design and development,
- responds correctly to all kinds of inputs,
- performs its functions within an acceptable time,
- is sufficiently usable,
- can be installed and run in its intended environments, and
- achieves the general result its stake holders desire.

As the number of possible tests for even simple software components is practically infinite, all software testing uses some strategy to select tests that are feasible for the available time (errors or other defects). Software testing can provide objective, independent information about the quality of software and risk of its failure to users and/or sponsors. Software testing can be conducted as soon as executable software (even if partially complete) exists. The overall approach to software development often determines when and how testing is conducted. For example, in a phased process, most testing occurs after system requirements have been defined and then implemented in testable programs. In contrast, under an Agile approach, requirements, programming, and testing are often done concurrently.

White-box Testing

Tests are performed to ensure that all internal operations of the software are performed according to the specifications of the client. This is called White box testing. White- box testing (also known as clear box testing, glass box testing, transparent box testing and structural testing) is a method of testing software that tests internal structures or workings of an application, as opposed to its functionality (i.e. black-box testing). In white- box testing an internal perspective of the system, as well as programming skills, are used to design test cases.

White-box testing can be applied at the unit, integration and system levels of the software testing process. Although traditional testers tended to think of white-box testing as being done at the unit level, it is used for integration and system testing more frequently today. It can test paths within a unit, paths between units during integration, and between subsystems during a system– level test. Though this method of test design can uncover many errors or problems, it has the potential to miss unimplemented parts of the specification or missing requirements.

Black-box Testing

Tests are performed to ensure that each function is working properly. This is referred to as Black-box testing. Black-box testing is a method of software testing that examines the functionality of an application (e.g. what the software does) without peering into its internal structures or workings. This method of test can be applied to virtually every

level of software testing: unit, integration, system and acceptance. It typically comprises most if not all higher level testing, but can also dominate unit testing as well. Test cases are built around specifications and requirements, i.e., what the application is supposed todo. Test cases are generally derived from external descriptions of the software, including specifications, requirements and design parameters. Although the tests used are primarily functional in nature, non-functional tests may also be used. The test designer selects both valid and invalid inputs and determines the correct output without any knowledge of the test object's internal structure. Testing is conducted in the system so that the functions namely Login, sending requests, searching the nearest donor, getting routes to the nearest blood banks etc... are done properly.

Condition Testing

Test cases are derived to determine whether the logic conditions and decision statements are free from errors. Condition testing strategy is used to check if the operators used are correct and to verify conditions such as if an error message is displayed if a non registered user is signed in to the app, or a user is registered without providing his body mass index value.

Loop Testing

This testing is used to check the variety of loops present in programming. The working of the loops such as while, for and do while are checked for its proper execution. The statements inside the loop body are executed line by line for every condition that satisfies the loop.

Unit Testing

Unit testing focused verification efforts on the smallest unit of software design, the module. This is also known as "module testing". The modules are tested separately. This testing is carried out during programming stage itself. In this testing step each module is found to be working satisfactorily as regard to the expected output from the module.

Integration Testing

Data can be lost across the interfaces; one module can have an adverse effect on the other; sub functions when combined, may not produce the desired major functions. The integration testing is a systematic testing for constructing the programs structure, while at the same time conducting tests to uncover errors associated within the interface. The objective is to take unit tested modules and build a program structure. All the modules are combines and tested as a whole. Here correction is difficult because the vast expenses of the entire program complicate the isolation of causes. Thus in the integration testing step, all the errors uncovered are corrected for the next testing steps.

Project aspect: Using integrated test plans prepared in the design phase of the system developed as a guide, the integration test was carried out. The modules are integrated and tested and all the errors found in the system were corrected for the next testing steps.

Output Testing

After performing the validation testing, the next step is output testing of the proposed system since no system could be useful if it does not produce the required output in specific format. Asking the users about the format required by them tests the outputs generated or displayed by the system under consideration. The output format of the screen is found to be correct as the format was designed in the system design phase according to the user needs. For the hard copy also, output comes out as the specified requirements by the user. Hence output testing does not result in any correction in the system. Various reports are generated in graphical output format and being pictorial representation it is found more convenient to understand by the users of the system.

User Acceptance Testing

User acceptance testing of a system is the key factor for the success of any system. The system under consideration is tested for user acceptance by constantly keeping in touch with the prospective system users at the time of developing and making changes wherever required. This is done with regard to the following pointsObjectives:

- A good case is one that has a high probability of finding an unpredictable error.
- A successful case is one that has a high probability of finding an unpredictable error.
- A good test case is one that provides solution to that unpredictable error.
- A test plan entailed the following activities. We prepare list plan.
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- Also we prepared list data transaction plan testing.
- Then we planned user training.
- Our programs were compiled and assembled.
- Job performance aids were prepared.

Test Results

Test results emphasize how the actual results differed from the expected results. This suggests the need for re-testing, and to discover the source of differences. The test phase of systems development process involves the defining of the criteria by which the system will be tested and measuring the criteria against the acceptable failure rate. Individual modules are tested during the development itself. Errors detected are corrected and re-tested, and the project leader has verified the compliance. Each input, output and processes are tested to verify that it performs as specified in the design. The units in the system are re-compiled and errors found are corrected as indicated by the compiler. The tests are repeated until all known errors are eliminated and the program matched the design specifications. Separate tests are performed to ensure that program units are properly interfaced with each other to form a complete system.

7.Future Enhancement

1. Advanced Artificial Intelligence

- Improve admission prediction accuracy using advanced machine learning models..
- Provide more personalized course and job recommendations based on student behavior and performance.

2. Real-Time Admission Tracking

- Allow students to track application status at every stage of the admission process.
- Provide instant notifications for approvals, rejections, and document verification.

3. Mobile Application Development

- Develop Android and iOS applications for better accessibility
- Enable students to receive alerts, attend tests, and view recommendations anytime.

4. Skill Assessment & Learning Analytics

- Introduce skill-based assessments to identify student strengths and weaknesses.
- Provide analytics dashboards to track performance and learning progress

5. Industry & Placement Integration

- Connect with companies to provide real-time job opportunities..
- Align academic programs with current industry requirements.

6. AI-Based Career Counseling

- Provide automated career guidance based on aptitude and interests.
- Suggest suitable career paths using job market trends

8.Conclusion

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The **Smart EduConnect System** successfully addresses the need for a centralized, intelligent, and automated educational platform that connects students, colleges, and administrators within a single digital ecosystem. By integrating advanced technologies such as Artificial Intelligence and data analytics, the system enhances decision-making processes and improves the efficiency of educational management.

The project provides a user-friendly environment where students can explore educational opportunities, receive personalized course and job recommendations, and predict admission chances based on their academic performance and interests. Colleges benefit from efficient data management, streamlined admission processes, and improved interaction with potential candidates, while administrators gain effective control over system operations, verification, and monitoring.

Through well-structured system design, secure data handling, and comprehensive testing, the Smart Edu Connect System ensures reliability, accuracy, and scalability. The implementation of features such as AI-powered chatbot assessments, real-time notifications, complaint management, and smart analytics demonstrates the system's ability to modernize traditional educational practices.

Overall, the Smart Edu Connect System proves to be a practical, scalable, and future-ready solution that simplifies educational processes, supports informed academic and career decisions, and contributes to the digital transformation of the education sector.

9.BIBLIOGRAPHY

Books

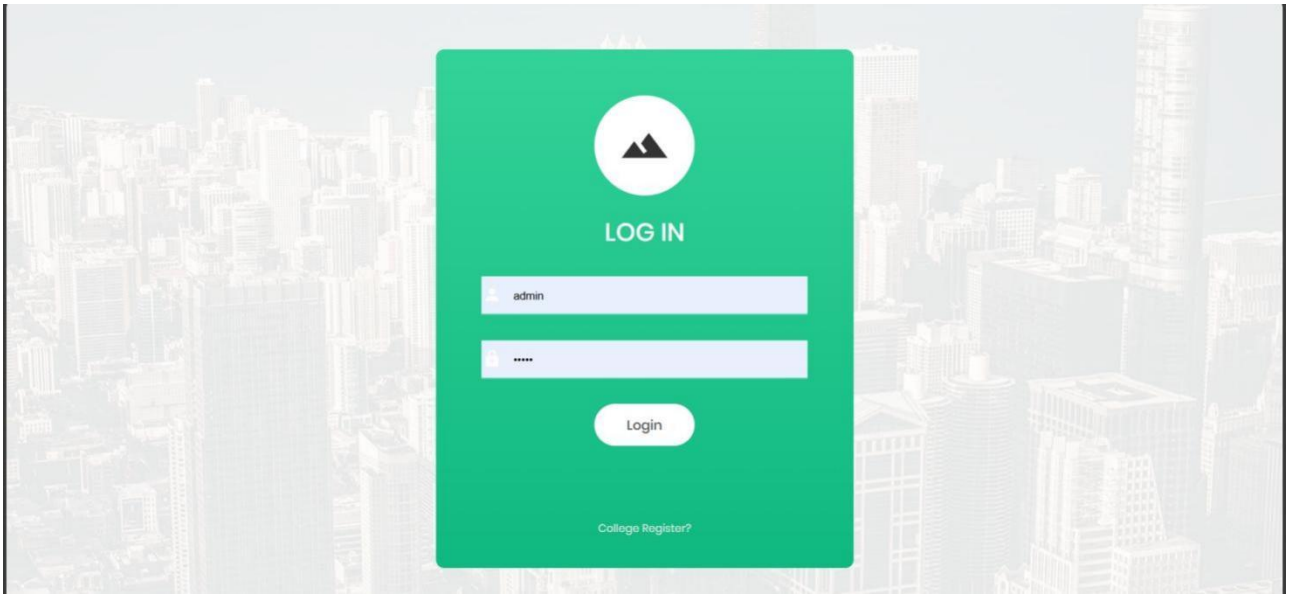
1. E. Balagurusamy, “Programming with Java: A Primer”
2. Brian W. Kernighan and Dennis M. Ritchie , “The C Programming Language”.
3. Marco L. Napoli, “Beginning Flutter: A Hands-On Guide to App Development”, Second Edition.

Links

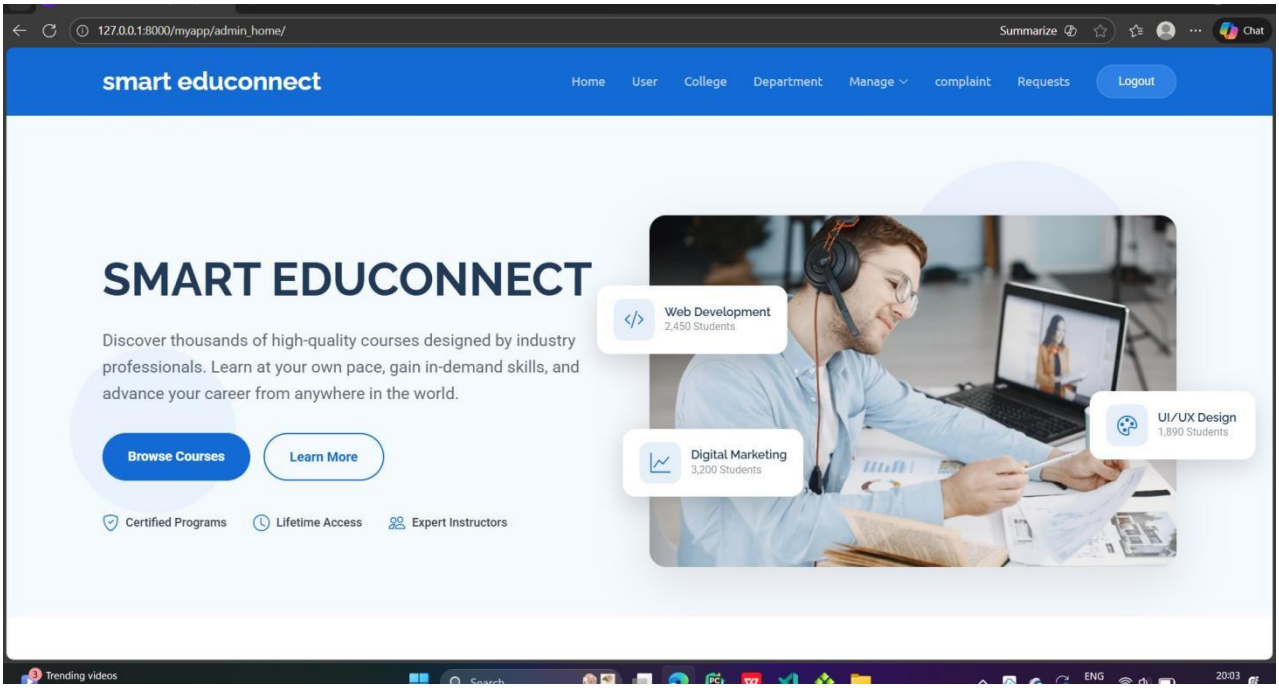
1. https://flutter.dev/
2. https://developer.android.com/studio
3. https://www.python.org/downloads/

10.Screenshot

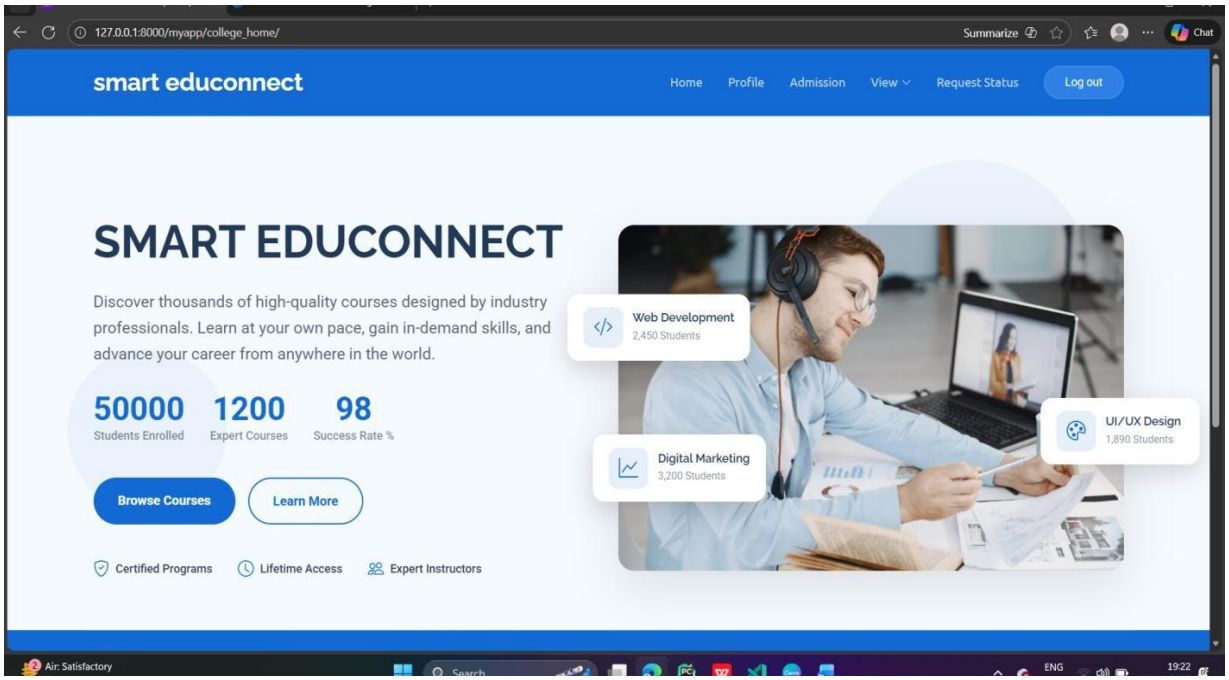
Login



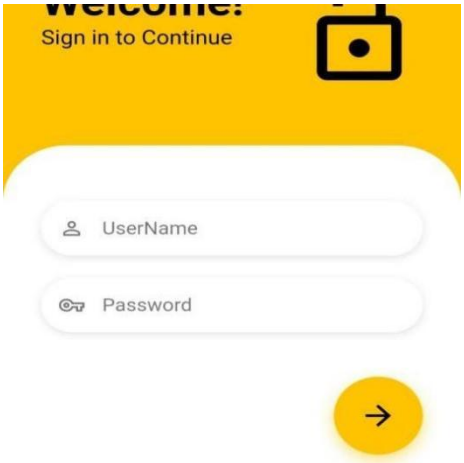
Admin



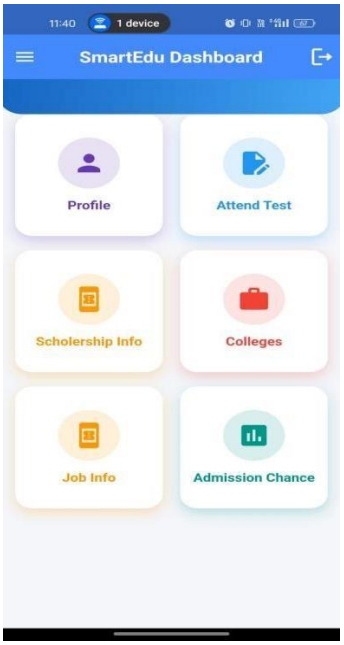
College



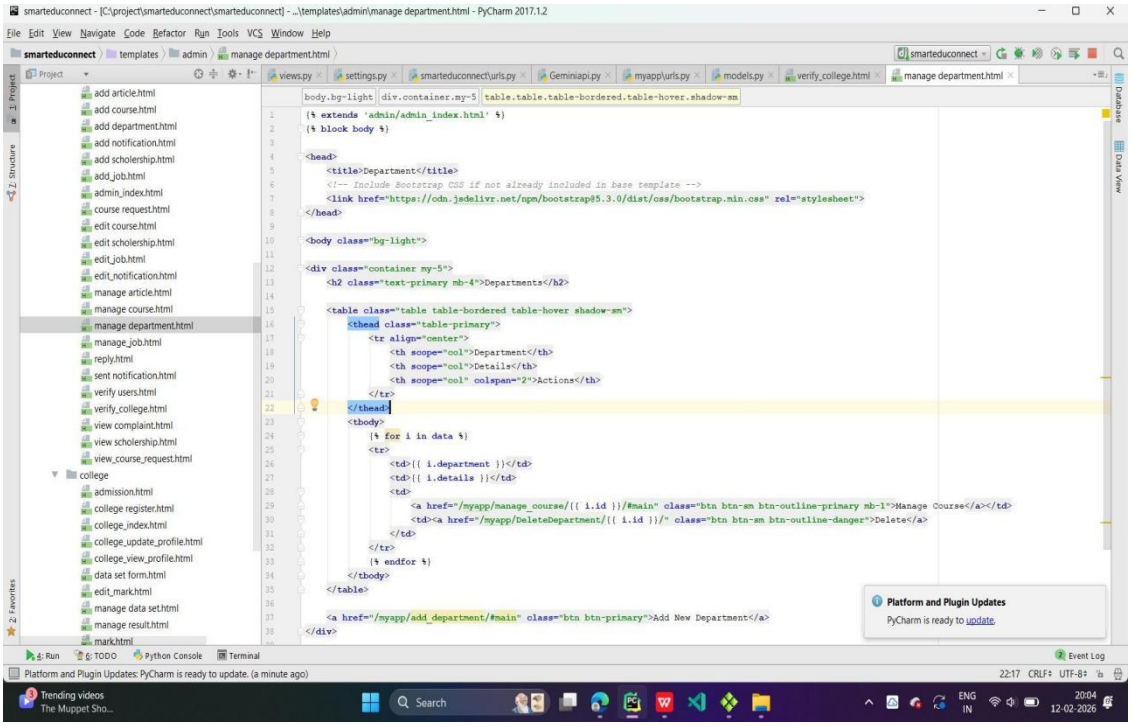
User Login



User



11.Source Code



```
{% extends 'admin/admin_index.html' %}
{% block body %}

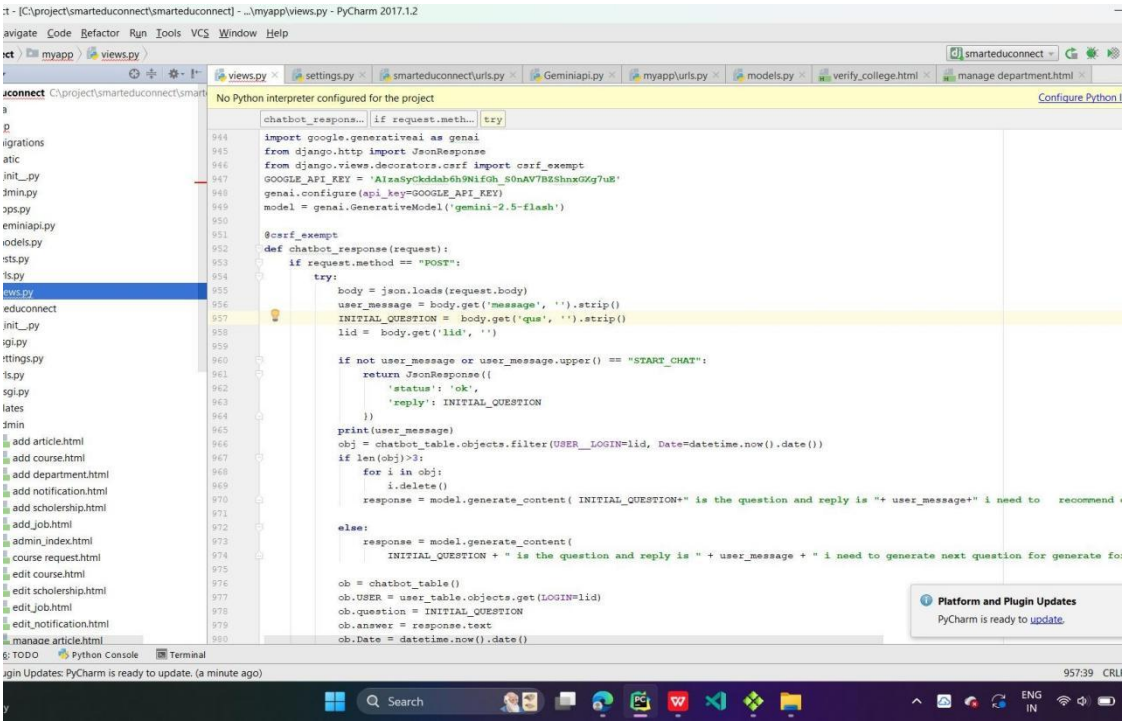
<head>
    <title>Department</title>
    <!-- Include Bootstrap CSS if not already included in base template -->
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
</head>

<body class="bg-light">

<div class="container my-5">
    <h2 class="text-primary mb-4">Departments</h2>

    <table class="table table-bordered table-hover shadow-sm">
        <thead class="table-primary">
            <tr align="center">
                <th scope="col">Department</th>
                <th scope="col">Details</th>
                <th scope="col" colspan="2">Actions</th>
            </tr>
        </thead>
        <tbody>
            {% for i in data %}
            <tr>
                <td>{{ i.department }}</td>
                <td>{{ i.details }}</td>
                <td>
                    <a href="/myapp/manage_course/{{ i.id }}/#main" class="btn btn-sm btn-outline-primary mb-1">Manage Course</a></td>
                    <td><a href="/myapp/DeleteDepartment/{{ i.id }}/" class="btn btn-sm btn-outline-danger">Delete</a>
                </td>
            </tr>
            {% endfor %}
        </tbody>
    </table>

    <a href="/myapp/add_department/#main" class="btn btn-primary">Add New Department</a>
</div>
```



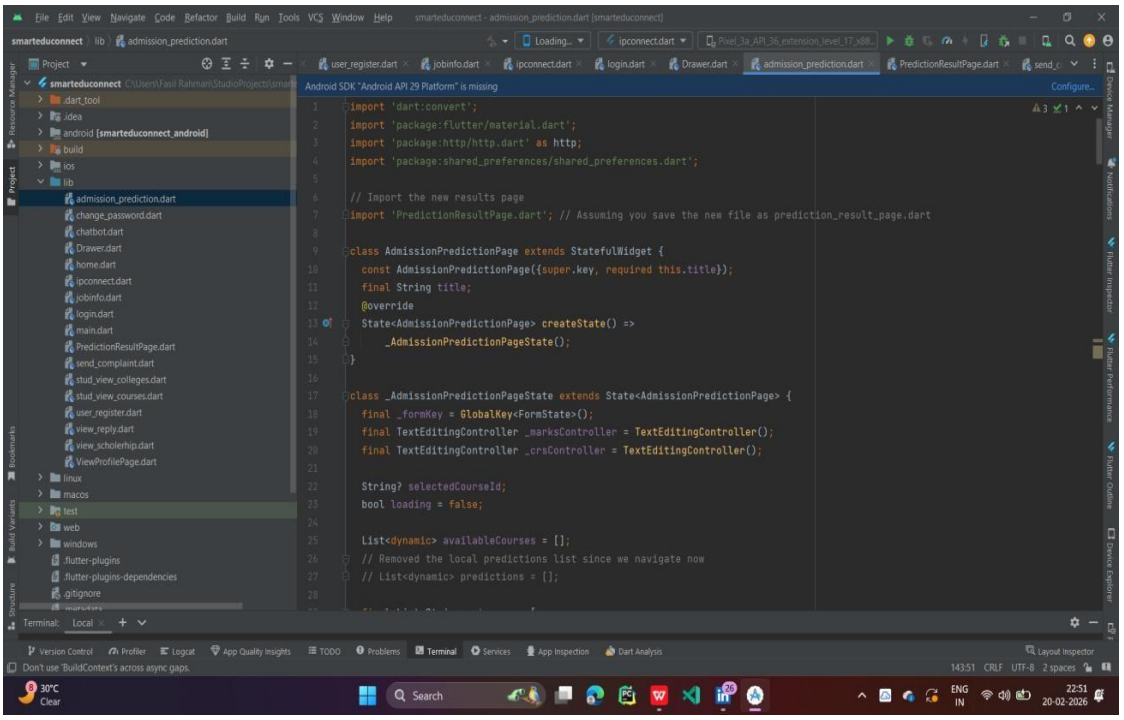
```
import google.generativeai as genai
from django.http import JsonResponse
from django.views.decorators.csrf import csrf_exempt
GOOGLE_API_KEY = 'AIzaSyCkddab6h9NifGh_S0nAV7BZShnxGXg7uE'
genai.configure(api_key=GOOGLE_API_KEY)
model = genai.GenerativeModel('gemini-2.5-flash')

@csrf_exempt
def chatbot_response(request):
    if request.method == "POST":
        try:
            body = json.loads(request.body)
            user_message = body.get('message', '').strip()
            INITIAL_QUESTION =  body.get('qus', '').strip()
            lid =  body.get('lid', '')

            if not user_message or user_message.upper() == "START_CHAT":
                return JsonResponse({
                    'status': 'ok',
                    'reply': INITIAL_QUESTION
                })
            print(user_message)
            obj = chatbot_table.objects.filter(USER__LOGIN=lid, Date=datetime.now().date())
            if len(obj)>3:
                for i in obj:
                    i.delete()
                response = model.generate_content( INITIAL_QUESTION+" is the question and reply is "+ user_message+" i need to   recommend

            else:
                response = model.generate_content(
                    INITIAL_QUESTION + " is the question and reply is " + user_message + " i need to generate next question for generate fo

            ob = chatbot_table()
            ob.USER = user_table.objects.get(LOGIN=lid)
            ob.question = INITIAL_QUESTION
            ob.answer = response.text
            ob.Date = datetime.now().date()
```



```dart
import 'dart:convert';
import 'package:flutter/material.dart';
import 'package:http/http.dart' as http;
import 'package:shared_preferences/shared_preferences.dart';

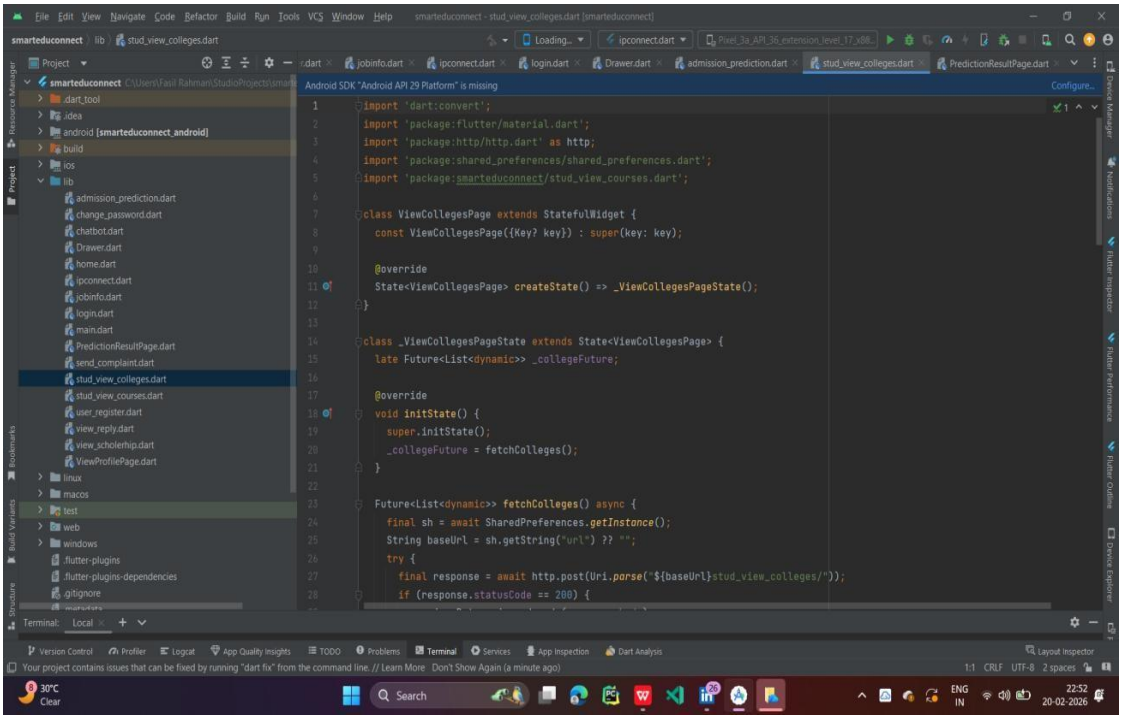
// Import the new results page
import 'PredictionResultPage.dart'; // Assuming you save the new file as prediction_result_page.dart

class AdmissionPredictionPage extends StatefulWidget {
  const AdmissionPredictionPage({super.key, required this.title});
  final String title;
  @override
  State<AdmissionPredictionPage> createState() =>
      _AdmissionPredictionPageState();
}

class _AdmissionPredictionPageState extends State<AdmissionPredictionPage> {
  final _formKey = GlobalKey<FormState>();
  final TextEditingController _marksController = TextEditingController();
  final TextEditingController _crsController = TextEditingController();

  String? selectedCourseId;
  bool loading = false;

  List<dynamic> availableCourses = [];
  // Removed the local predictions list since we navigate now
  // List<dynamic> predictions = [];
```



```dart
import 'dart:convert';
import 'package:flutter/material.dart';
import 'package:http/http.dart' as http;
import 'package:shared_preferences/shared_preferences.dart';
import 'package:smarteduconnect/stud_view_courses.dart';

class ViewCollegesPage extends StatefulWidget {
  const ViewCollegesPage({Key? key}) : super(key: key);

  @override
  State<ViewCollegesPage> createState() => _ViewCollegesPageState();
}

class _ViewCollegesPageState extends State<ViewCollegesPage> {
  late Future<List<dynamic>> _collegeFuture;

  @override
  void initState() {
    super.initState();
    _collegeFuture = fetchColleges();
  }

  Future<List<dynamic>> fetchColleges() async {
    final sh = await SharedPreferences.getInstance();
    String baseUrl = sh.getString("url") ?? "";
    try {
      final response = await http.post(Uri.parse("${baseUrl}stud_view_colleges/"));
      if (response.statusCode == 200) {
```